

LB-AO v0.1

Formal Pelvis & Hip Complex Ontology
Musculoskeletal Digital Twin Project

Purpose	Create a stable, shareable anatomical reference layer for the pelvis and hip complex that remains more detailed than any initial simulation abstraction.
Core principle	Anatomical fact, biomechanical abstraction, and computational implementation are recorded separately.
Initial application	Support the lower-body sagittal squat model while preserving a path toward 3-D, muscle-driven, subject-specific digital twins.
Ontology ID	LB-AO = Lower-Body Anatomy Ontology. Version 0.1 establishes the Pelvis & Hip Complex module.

Reference Philosophy

The ontology is intentionally richer than the first squat simulation. A simplified model may represent the pelvis as one rigid body and the hip as a single sagittal rotational coordinate, while the reference ontology retains the actual bony, articular, passive-tissue, muscular, coordinate-system, and modeling relationships needed for future fidelity.

REFERENCE ANATOMY -> BIOMECHANICAL REPRESENTATION -> COMPUTATIONAL IMPLEMENTATION

1. Identifier and Class System

Human-readable, persistent identifiers allow the same entity to be referenced later from YAML, Python objects, simulation files, measurements, and research notes.

Prefix	Meaning
REG	Anatomical region
BONE	Bone / osseous structure
LMK	Anatomical or derived landmark
SURF	Articular surface
JOINT	Joint / articulation
CART	Articular cartilage
LAB	Labrum
CAP	Joint capsule
LIG	Ligament
MUS	Muscle
TEN	Tendon / aponeurotic structure
NERVE	Nerve
SEG	Biomechanical segment
FRAME	Reference / coordinate frame
DOF	Modeled degree of freedom
MODEL	Computational abstraction

Examples: **LB:BONE:ILIUM**, **LB:JOINT:HIP_R**, **LB:MUS:GLUTEUS_MAXIMUS**, **LB:FRAME:PELVIS**.

2. Top-Level Knowledge Structure

The module is organized by anatomical entity type rather than by one simulation's needs. This lets one entity participate in several systems without duplication.

```
LB:REG:PELVIS_HIP_COMPLEX
|
+-- Skeletal anatomy
+-- Articular anatomy
+-- Passive stabilizers
+-- Musculotendon system
+-- Neuromuscular relationships
+-- Biomechanical representation
+-- Computational abstraction
```

3. Pelvic Skeletal Complex

The bony pelvis includes the two hip bones together with the sacrum and coccyx. Each hip bone contains ilium, ischium, and pubis regions; these converge around the acetabulum. Mechanically, the pelvis transfers load between the trunk and lower limbs and supplies extensive attachment sites.

```

LB:BONE:PELVIS
+-- LB:BONE:HIP_BONE_R
| +-- ilium
| +-- ischium
| +-- pubis
+-- LB:BONE:HIP_BONE_L
+-- LB:BONE:SACRUM
+-- LB:BONE:COCCYX

```

4. Hip Bone and Essential Landmarks

Landmarks are explicit entities because they can define coordinate frames, locate muscle and ligament attachments, support motion-capture registration, and anchor subject-specific geometry.

```

ILIUM
ASIS | AIIS | PSIS | PIIS | iliac crest
gluteal surface | greater sciatic notch
ISCHIUM
ischial tuberosity | ischial spine
lesser sciatic notch | ischial ramus
PUBIS
pubic body | superior/inferior pubic rami
pubic crest | pubic tubercle

```

5. Acetabulum and Proximal Femur

The femoral head articulates with the acetabulum. Their osseous geometry contributes strongly to stability; the labrum further deepens the socket and contributes to joint sealing and stability.

```

LB:SURF:ACETABULUM
PART_OF -> hip bone
FORMED_BY_REGIONS_OF -> ilium, ischium, pubis
ARTICULATES_WITH -> femoral head
ASSOCIATED_WITH -> labrum, cartilage, capsule

PROXIMAL FEMUR
femoral head | fovea capitis | femoral neck
greater trochanter | lesser trochanter

```

6. Hip Joint Ontology

The anatomical hip is a multiaxial synovial ball-and-socket joint. A sagittal squat model may retain only flexion/extension, but that is stored as a model assumption rather than an anatomical claim.

```

LB:JOINT:HIP
CONNECTS -> pelvis <-> femur
ANATOMICAL MOTIONS
flexion / extension
abduction / adduction
internal / external rotation
SQUAT MODEL v0.1
RETAINS -> flexion/extension
OMITS -> explicit frontal/transverse rotations

```

7. Cartilage, Labrum, and Capsule

Articular cartilage supports load distribution and low-friction articulation. The acetabular labrum is fibrocartilaginous, deepens the socket, and contributes to the fluid seal and stability. The fibrous capsule surrounds the joint and is reinforced by major capsular ligaments.

```

HIP ARTICULAR CARTILAGE
COVERED_BY relationship on opposing surfaces
ACETABULAR LABRUM
ATTACHES_TO -> acetabular rim

```

```
HIP_CAPSULE
SURROUNDS -> articulation
REINFORCED_BY -> capsular ligaments
```

8. Ligament System

The iliofemoral, pubofemoral, and ischiofemoral ligaments reinforce the capsule and contribute passive restraint. Ligament mechanics can later be represented with configuration-dependent length, strain, and force.

```
CAPSULAR
LB:LIG:ILIOFEMORAL
LB:LIG:PUBOFEMORAL
LB:LIG:ISCHIOFEMORAL
zona orbicularis
INTRACAPSULAR
LB:LIG:FEMORAL_HEAD
PELVIC_STABILIZERS
sacroiliac ligaments
sacrospinous | sacrotuberous
```

9. Musculotendon Inventory

A muscle is represented by attachments, path geometry, joints crossed, innervation, and musculotendon properties. Functional actions emerge from force and geometry rather than being the only stored description.

```
GLUTEAL
gluteus maximus | medius | minimus | TFL
ILIOPSOAS
psoas major | iliacus
ADDUCTOR / MEDIAL
adductor magnus | longus | brevis
pectineus | gracilis
HAMSTRING-RELATED HIP SPANNERS
semimembranosus | semitendinosus
biceps femoris long head
DEEP EXTERNAL ROTATORS
piriformis | obturators | gemelli | quadratus femoris
OTHER BIARTICULAR
rectus femoris | sartorius
```

10. Formal Muscle Pattern - Gluteus Maximus

This pattern is intended to be reusable for every lower-body muscle. Detailed attachment sites and numerical model parameters should be separately sourced and versioned rather than embedded as undocumented constants.

```
LB:MUS:GLUTEUS_MAXIMUS
CLASS -> skeletal muscle
CROSSES -> hip joint
MODEL_PARAMETERS
max isometric force
optimal fiber length
tendon slack length
pennation angle
activation/deactivation dynamics
DYNAMIC_VARIABLES
excitation(t), activation(t)
fiber/tendon state, muscle force(t)
```

11. Moment Arm and Force Transmission

Moment arm is configuration-dependent and belongs explicitly in the biomechanical layer. This prevents a muscle from being represented as if it simply stores a fixed joint torque.

```

Muscle -> GENERATES -> MuscleForce
-> ACTS_THROUGH -> MomentArm(q)
-> ABOUT -> HipJoint
-> PRODUCES -> JointMomentContribution

tau_i(q) = r_i(q) * F_i

```

12. Neuromuscular Layer

Motor innervation belongs in the reference ontology, while detailed neural circuitry can remain outside the first simulation. This separates anatomical connectivity, physiological activation dynamics, and mechanical force production.

```

Nerve -> INNERVATES -> Muscle
Muscle -> RECEIVES -> NeuralExcitation
Excitation -> DRIVES -> Activation
Activation -> MODULATES -> MuscleForce
MuscleForce -> CONTRIBUTES_TO -> Joint mechanics

```

13. Coordinate Frames and Joint Center

Coordinate frames are modeling and measurement constructs, not anatomical structures. The ISB recommends standardized local axes and joint-coordinate definitions for reporting hip motion; OpenSim models likewise use explicit segment-fixed frames.

```

LB:FRAME:PELVIS
ATTACHED_TO -> pelvis segment
DEFINED_FROM -> documented landmarks/convention
LB:FRAME:FEMUR
ATTACHED_TO -> thigh/femoral segment
HIP JOINT CENTER
TYPE -> derived / functional-geometric landmark
USED_BY -> kinematics and joint geometry

```

14. Biomechanical Abstraction

Model reduction is explicit. Multiple anatomical structures may be represented by one rigid body while their identities and relationships remain in the reference ontology.

```

ANATOMICAL PELVIS
hip bones + sacrum + coccyx + joints + ligaments
|
| REPRESENTED_BY
v
LB:SEG:PELVIS (RigidBody)
mass | COM | inertia | pose | velocity

ANATOMICAL FEMUR
|
v
LB:SEG:THIGH (RigidBody)

```

15. Squat v0.1 Mapping

The first squat model is a view of the ontology, not the ontology itself. Increasing fidelity should change the representation layer without redefining underlying anatomy.

```

REFERENCE KNOWLEDGE
acetabulum + femoral head + labrum + cartilage
capsule + ligaments + muscles + 3-D joint motion
|
v
BIOMECHANICAL ABSTRACTION
rigid pelvis -- idealized hip -- rigid thigh
|

```

v
COMPUTATIONAL MODEL
pelvis segment + hip flex/ext coordinate + thigh

16. Controlled Relationship Vocabulary

Relationship	Meaning
IS_A	Class/subclass
PART_OF / HAS_PART	Anatomical composition
ATTACHES_TO	Anatomical attachment
ARTICULATES_WITH	Joint/surface articulation
CROSSES	Structure spans a joint
CONNECTS	Structural connection
COVERED_BY	Surface covering
INNERVATED_BY	Neural relationship
GENERATES	Produces a physical quantity
TRANSMITS_FORCE_TO	Force pathway
ACTS_ON	Mechanical influence
CONSTRAINS	Passive restriction
STABILIZES	Contribution to stability
HAS_FRAME	Reference-frame association
REPRESENTED_BY	Anatomy-to-model abstraction
MEASURED_BY	Direct observation
ESTIMATED_BY	Derived/estimated quantity

17. Machine-Readable Entity Schema

Every entity should share a schema that keeps anatomical evidence, model choices, and assumptions inspectable.

```
id: LB:JOINT:HIP_R
preferred_name: Right Hip Joint
entity_class: SynovialJoint
anatomical_properties:
  joint_type: ball_and_socket
  multiaxial: true
relationships:
  articulates:
    - right_acetabulum
    - right_femoral_head
  computational_representation:
    squat_v0.1:
      type: idealized_joint
      active_coordinate:
        - hip_flexion_extension
  assumptions:
    - sagittal_plane_model
    - rigid_pelvis
    - rigid_thigh
    - idealized_joint_center
    - no_explicit_cartilage_contact
  evidence:
```

- source_id
- source_version
- retrieval_date

18. Accuracy and Evidence Policy

Anatomical facts should be linked to authoritative references. Numerical biomechanical parameters should additionally record units, source population, coordinate convention, and model assumptions. Model-specific values must not silently become universal anatomical facts.

For hip kinematics, standardized coordinate conventions such as the International Society of Biomechanics recommendations should be preferred or explicitly mapped. OpenSim is a useful example of translating anatomy into computational bodies, frames, joints, and musculotendon paths, but those choices remain model representations.

19. Reference Sources

Reference	Use in this framework
NCBI Bookshelf / StatPearls: Anatomy, Bony Pelvis and Lower Limb, Hip	Hip articulation, osseous stability, capsule, capsular ligaments, labrum, cartilage, and functional context.
NCBI Bookshelf / StatPearls: Anatomy, Bony Pelvis and Lower Limb, Hip Joint	Hip-joint structure, axes of motion, capsular ligaments, ligamentum teres, and stability.
OpenStax Anatomy & Physiology 2e	Pelvic/hip-bone organization and hip-joint anatomy.
Wu et al., Journal of Biomechanics (2002), ISB Recommendation	Standardized joint-coordinate-system recommendations for hip kinematics.
OpenSim Gait2392 / Gait2354 documentation	Example segment-fixed reference frames and computational lower-extremity abstractions.

20. Next Project Artifact

The next implementation artifact should be a YAML version of this module using the stable IDs and controlled relationships defined here. We can validate the schema on the pelvis/hip complex before propagating it to the thigh, knee, leg, ankle, and foot.